

QEM-8³: Hierarchical Mirror Lattice and Sequential Flip-Wave Dynamics in Multi-Body Entanglement

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Abstract: We present QEM-8³, a hierarchical mirror-lattice framework for multi-body quantum entanglement. Non-commutativity of measurement order and basis rotation jointly induce Sequential Trigger Flips (STF), whose propagation forms Flip-Waves across nested mirror layers. We derive a wave equation coupling sequence states and their mirror counterparts, identify a Critical Layer (CL) at mirror-basis resonance, and define the symmetry group $G_{\{8^3\}} = U(1)_{\text{phase}} \times SU(2)_{\text{flip}} \times Z2_{\text{mirror}} \times Z3_{\text{hierarchy}}$ together with a topological invariant $T_{\{8^3\}}$. Predictions include quasi-time-crystalline spectral oscillations, time-reversal echoes, and critical entanglement peaks.

Keywords: hierarchical entanglement; mirror symmetry; measurement order; flip-wave; critical layer; topological invariant

Note

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